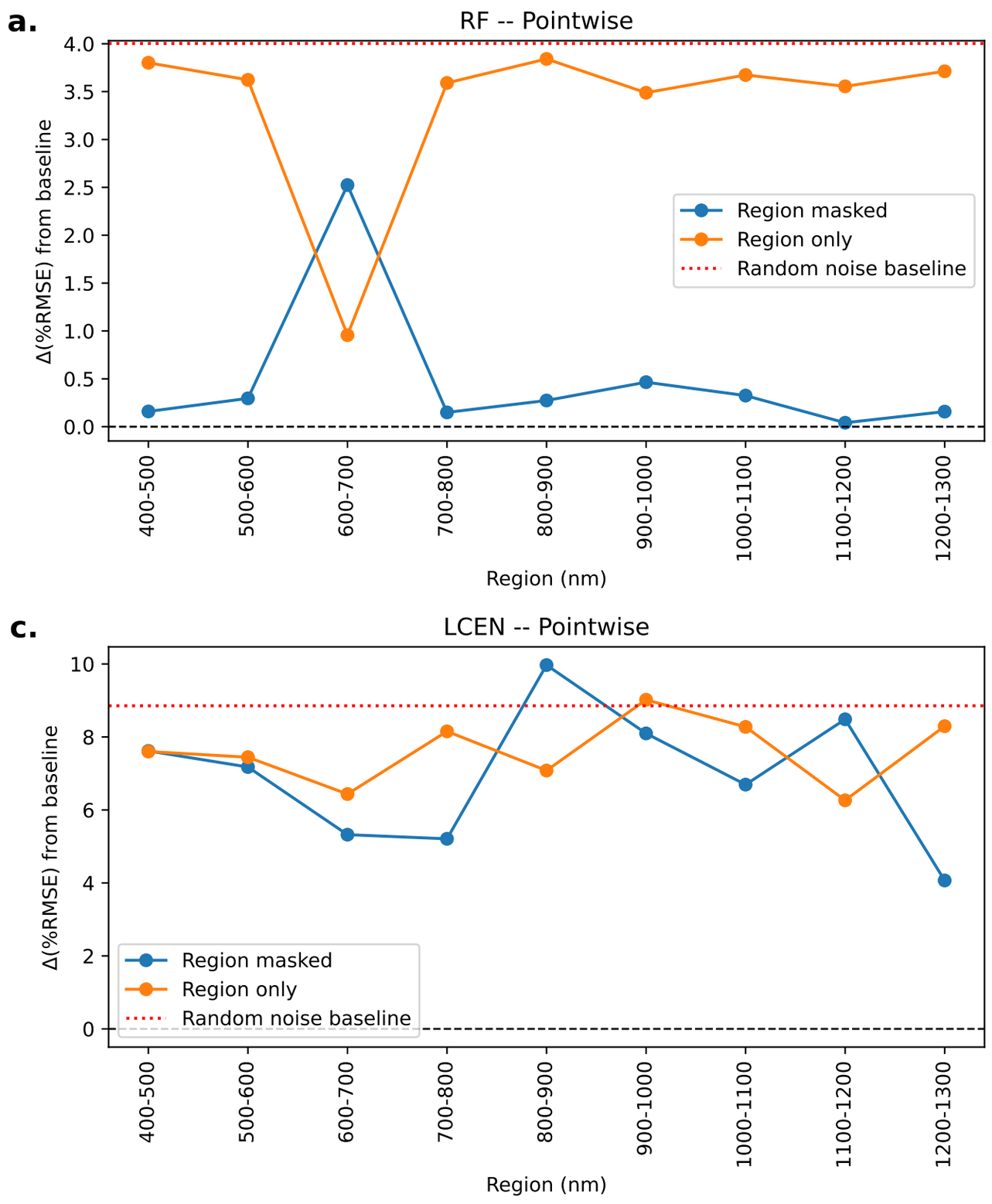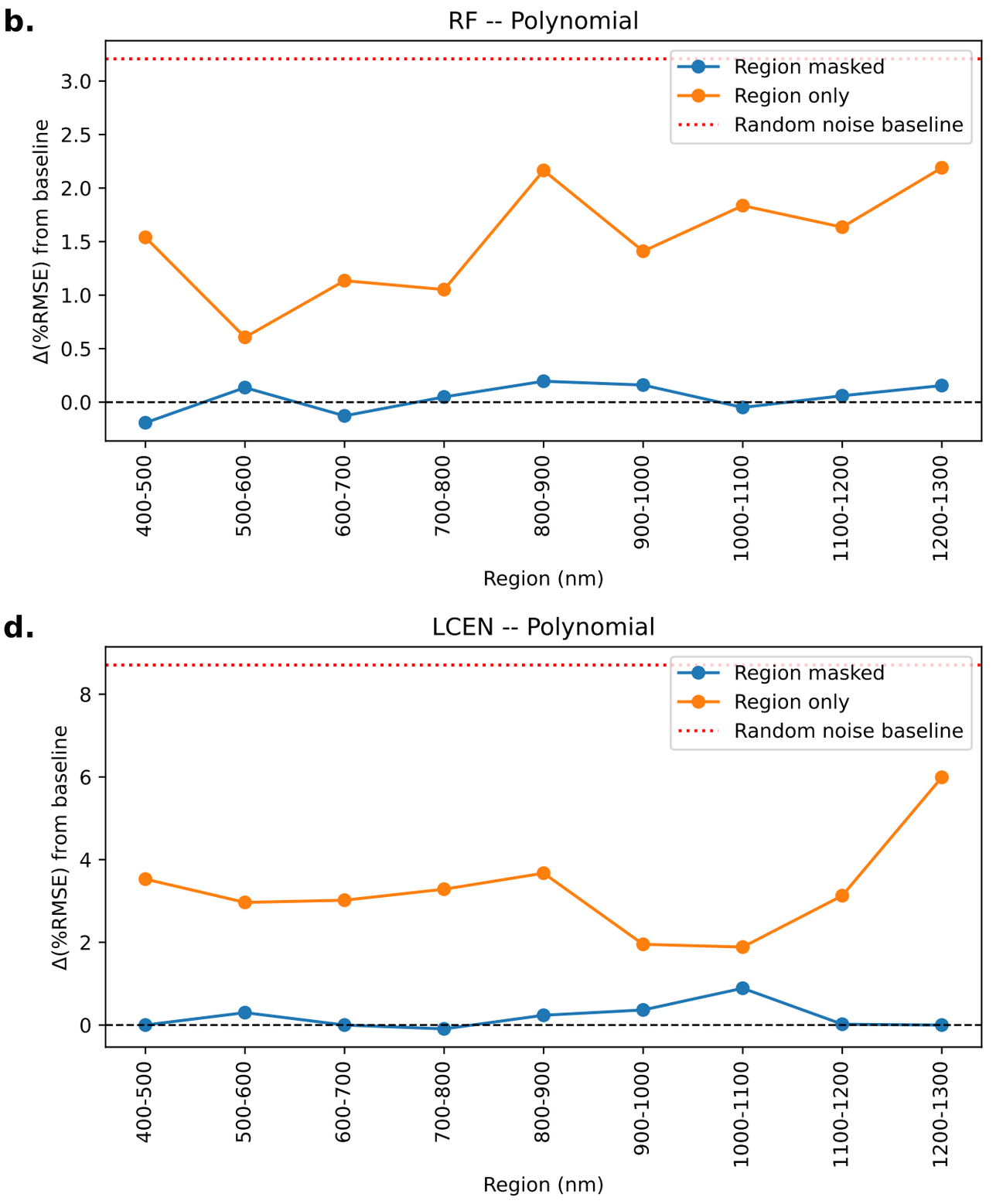Region importance via noise masking